Topic 3: Data

Computers are able to store and manipulate large quantities of data. They use binary to represent different types of data.

Students are expected to learn how different types of data are represented in a computer.

	Students should:
3.1 Binary	**3.1.1** Understand that computers use binary to represent data (numbers, text, sound, graphics) and program instructions. **3.1.2** Understand how computers represent and manipulate numbers (unsigned integers, signed integers (sign and magnitude, two's complement)). **3.1.3** Be able to convert between binary and denary whole numbers (0–255). **3.1.4** Understand how to perform binary arithmetic (add, shifts (logical and arithmetic)) and understand the concept of overflow. **3.1.5** Understand why hexadecimal notation is used and be able to convert between hexadecimal and binary.